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Windshield for a motor vehicle

Abstract

A windshield for a motor vehicle is described, comprising: a frame with a first rim defining a first opening, and a first window engaging the first opening, supported by the first rim; the frame comprises, in turn, a first and a second cross member opposite one another; and one pair of first pillars opposite one another and extending crosswise to the first and second cross member; the first and second cross member and said pillars delimit said first rim; the frame comprises a second rim and a third rim distinct from one another and arranged on respective opposite sides of the first rim; the frame comprises a pair of third cross members smoothly joined to the first cross member by respective opposite side bands of the first cross member; and a pair of fourth cross members smoothly joined to the second cross member; the second rim is delimited by the first pillar, by a respective third cross member and by a respective fourth cross member; the third rim is delimited by the second pillar, by a respective third cross member and by a fourth cross member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application claims priority from Italian patent application no. 102022000003431 filed on Feb. 24, 2022, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) This invention relates to a windshield for a motor vehicle and to a motor vehicle comprising this windshield.

BACKGROUND

(3) Motor vehicles comprising a body defining a passenger compartment and a windshield arranged at the front of the passenger compartment are known.

(4) Sports cars with an uncovered passenger compartment defining a frame and a windshield shaped like a dome fixed to the frame have also been developed in the past.

(5) In particular, the dome-shaped windshield extends wholly both at the front and to the side of the passenger compartment so as to improve the aerodynamic properties of the race car.

(6) Being intended for race cars, the use of a dome-shaped windshield did not need to meet compliance, assembly safety, or aesthetic constraints.

(7) In the sector, there is the need to use, for stylistic reasons, dome-shaped windshields on standard motor vehicles, meeting the typical technical requirements of standard motor vehicles.

(8) In particular, there is the need in the sector to produce the dome-shaped windshield so as to enable adequate visibility from inside the passenger compartment and to enable its shattering in the event of a collision.

(9) In addition, the need is felt in the sector to glue the windshield to the frame using a bead of adhesive material without letting the adhesive material fall, or limiting this.

SUMMARY

(10) The purpose of this invention is to produce a windshield for a motor vehicle, which makes it possible to meet the need mentioned above.

(11) The above-mentioned purpose is achieved with this invention, as it relates to a windshield for a motor vehicle according to what is defined in claim 1.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to better understand this invention, a non-limiting preferred embodiment thereof will now be described by way of example with reference to the accompanying drawings, in which:

(2) FIG. 1 is a perspective view of a motor vehicle with a windshield produced according to the precepts of this invention;

(3) FIGS. 2 to 4 are respective cross-sections on a greatly enlarged scale of the windshield in FIG. 1 according to the respective lines II-II; III-III and IV-IV.

DESCRIPTION OF EMBODIMENTS

(4) With reference to the attached figures, reference number 1 indicates a motor vehicle comprising a body 2 defining a passenger compartment 3.

(5) The motor vehicle 1 is, in the case illustrated, a standard motor vehicle.

- (6) It should be specified that, below in this description, expressions such as “above”, “below”, “in front of”, “behind” and the like are used with reference to the normal movement of the motor vehicle **1**.
- (7) It should also be specified that, below in this description, the expression “concave” (“convex”) region is used to indicate a two-dimensional or three-dimensional region shaped so that at least one segment exists (or none exist) joining two points of the region that does not entirely belong to the same region.
- (8) It should, finally, be specified that below in this description the expression segment with an upwards (“downwards”) concavity is used to indicate a segment or a single-dimensional curve shaped so as to be located above (below) any tangent to the segment or curve itself.
- (9) The passenger compartment **3** is delimited at the front by a windshield **4** and behind by a backrest **5**.
- (10) The passenger compartment **3** is also delimited above by a roof **6**.
- (11) The roof **6** extends between the backrest **5** and the windshield **4**.
- (12) The motor vehicle **1** comprises, in addition, a pair of doors **7** hinged to the body **2** and moveable between: a closed position in which they delimit, to the side, the passenger compartment **3** and prevent entry into/exit from the passenger compartment **3**; and an open position in which they enable entry into/exit from the passenger compartment **3**.
- (13) It is also possible to define: a longitudinal axis X integral with the motor vehicle **1**, arranged, in use, horizontal and parallel to a normal direction of travel of the motor vehicle **1**; a transversal axis Y integral with the motor vehicle **1**, arranged, in use, horizontal and orthogonal to the axis X; and an axis Z integral with the motor vehicle **1**, arranged, in use, vertical and orthogonal to the axes X, Y.
- (14) More specifically, the windshield **4** comprises, in turn: a frame **10** comprising a rim **11** defining an opening **12**; and a front window **19** engaging the opening **12** supported by the rim **11** and arranged in front in relation to a normal forward moving direction parallel to the direction X.
- (15) More specifically, the rim **11** is quadrangular shaped and comprises: a pair of upper and lower cross members **12**, **13** opposite to each other; and a pair of side pillars **14**, **15** opposite one another and extending crosswise between corresponding ends of the cross members **12**, **13**.
- (16) The cross members **12**, **13** have an extension mainly parallel to the axis Y.
- (17) The first cross member **13** is arranged behind the second cross member **12**.
- (18) Advantageously, the frame **10** comprises a pair of rims **20** distinct from one another and arranged on respective opposite sides of the rim **10**; the rims **20** delimit respective openings **25**; the windshield **4** comprises, in addition, a pair of side windows **26** engaging respective openings **25**, supported, respectively, by the rims **20** of the frame **10**, and arranged to the side with reference to the normal forward moving direction of the motor vehicle **1**;
- (19) the frame **10** comprises, in addition: a pair of cross members **22** smoothly joined to the cross member **12** and arranged by respective opposite side bands of the cross member **12**; and a pair of cross members **21** smoothly joined to the cross member **13** and arranged by respective opposite side bands of the cross member **13**;
- (20) each rim **20** is delimited by a respective pillar **14**, **15**, by a respective cross member **12**, and by a respective second cross member **13**.
- (21) In particular, the windows **19**, **26** are tangent to each other.
- (22) The cross members **21**, **22** are joined to the corresponding cross members **12**, **13** without forming sharp edges.
- (23) The cross members **21**, **22** of each rim **20** comprise: first ends **23** connected to each other; and second ends connected to a corresponding end of the cross member **12** and a corresponding end of the second cross member **13**.
- (24) The cross members **21**, **22** of each rim **20** preferably form a straight angle having a vertex on the corresponding end **23**.

- (25) Each rim **20** is triangular.
- (26) The rims **11** and **20** are concave.
- (27) The cross member **12** has a concavity facing downwards.
- (28) The cross member **13** has a concavity facing backwards. a ring-shaped gasket **60** surrounding the window **15** and interposed between the window **19** and the rim **11** (FIGS. **3** and **4**); and a pair of gaskets **65** comolded on the windows **26** (only one of which is illustrated in FIG. **1**).
- (29) The window **19** is glued to the rim **11**. The windows **26** are glued to the corresponding rims **20**.
- (30) More specifically, the window **19** is glued to the pillars **14, 15** using a bead **71** of adhesive (FIG. **2**).
- (31) The windows **26** are glued to the pillar **14, 15** using a bead **75** of adhesive (FIGS. **2** and **4**).
- (32) With reference to FIG. **3**, the window **19** comprises an upper edge **61** abutting against the gasket **60** and glued to the cross member **12** using an adhesive bead **70**.
- (33) With reference to FIG. **2**, the window **19** comprises a side edge **62** arranged so it abuts against the gasket **65** and glued to the pillar **14** using the bead **71**.
- (34) Each gasket **65** is, in addition, comolded on a side end edge **66** of the corresponding window **19**.
- (35) With reference to FIG. **4**, the gasket **60** and each gasket **65** are arranged at corresponding adjacent sides of the corresponding pillar **14, 15**.
- (36) In particular, the windows **26** are made of tempered glass.
- (37) The window **15** is made of a multi-layer material formed from a pair of laminated glass plates between which a film of plastic material, especially polyvinyl butyral, is interposed.
- (38) The assembly of the windshield **4** occurs by interposing the gasket **60** between the window **19** and the rim **11**, and gluing the window **19** to the rim **11** using the beads **70, 71**.
- (39) The windows **26** with the corresponding comolded gaskets **65** are then glued to the pillars **14** using respective beads **75**, until interposing each gasket **65** between the window **19** and the corresponding window **26**.
- (40) In particular, the concave windows **19, 26** are brought near the frame **10** after the application of the adhesive forming the beads **70, 71, 75** using a movement perpendicular to the frame **10** that compresses the beads **70, 71, 75** and limits—or even eliminates—the risk of their falling from the frame **10**.
- (41) The windshield **4** comprises, in one embodiment, multiple accessories (not illustrated in the attached figures), in particular a rear-view mirror and/or a roof lamp, fixed to the windshield **19**.
- (42) The frame **10** is made of carbon fibre.
- (43) The advantages enabled by the windshield **4** according to this invention will be apparent from an examination thereof.
- (44) In particular, the frame **10** defines both the rim **11** for the window **19** and the rims **20** for the corresponding windows **26**.
- (45) In this way, it is possible to produce a dome-shaped frame **10**, meeting, at the same time, the legal and technological requirements typical of the standard motor vehicle **1**.
- (46) This enables, on the one hand, the production of the window **19** in a multi-layer material that ensures correct visibility from the inside of the passenger compartment **3** and limits the risk of shattering in the event of a collision, and, on the other hand, the production of the windows **26** made of tempered glass since the latter do not need to meet stringent requirements regarding visibility and limiting the risk of shattering.
- (47) Thanks to the fact that the windows **19, 26** are separate from each other, it is possible to bring the windows **19, 26** near the frame **10** after having applied the beads **70, 71, 75** using a movement that is basically perpendicular to the beads **70, 71**. It is, thus, possible to glue the concave windows **19, 26**, limiting the risk of the adhesive substance's sliding from the frame **10**.
- (48) The gaskets **60, 65** are interposed between the window **19** and the windows **26**, preventing

direct contact thereto. In this way, the fluid tightness is significantly improved.

(49) The gaskets **65** are comolded on the corresponding windows **26**, simplifying the assembly thereof and improving the finish and geometric precision thereof.

(50) Finally, it is clear that changes may be made to the windshield **4**, and variations produced thereto, according to this invention that, in any case, do not depart from the scope of protection defined by the claims.

Claims

1. A windshield (**4**) for a motor vehicle (**1**), comprising: a frame (**10**) comprising a first rim (**11**), which defines a first opening (**15**); and a first window (**19**) engaging said first opening (**15**), supported by said first rim (**11**) and arranged at the front, with reference to a normal forward moving direction of said motor vehicle (**1**); said frame (**10**) comprising, in turn: a first and a second cross member (**12, 13**) opposite one another and extending, in use, crosswise to said forward moving direction; and a pair of first pillars (**14, 15**) opposite one another and extending crosswise to said first and second cross member (**12, 13**); said first and second cross member (**12, 13**) and said pillars (**14, 15**) delimiting said first rim (**11**); wherein said frame (**10**) further comprises a second rim and a third rim (**20**) distinct from one another and arranged on respective opposite sides of said first rim (**10**); said second and third rim (**20**) delimiting a second opening and a third opening (**25**), respectively; said windshield (**4**) further comprising: a second and a third window (**26**) engaging said second and third opening (**25**), respectively, supported by said second and third rim (**20**) of said frame (**10**), respectively, and arranged laterally with reference to said normal forward moving direction of said motor vehicle (**1**); said frame (**10**) further comprising: a pair of third cross members (**22**) smoothly joined to said first cross member (**12**) by respective opposite side bands of said first cross member (**12**); a pair of fourth cross members (**21**) smoothly joined to said second cross member (**13**) by respective opposite side bands of said second cross member (**13**); said second rim (**20**) being delimited by said first pillar (**14**), by a respective third cross member (**22**) and by a respective fourth cross member (**21**); said third rim (**20**) being delimited by said second pillar (**15**), by a respective third cross member (**22**) and by a respective fourth cross member (**21**); a first gasket (**60**) shaped like a ring fitted onto said first window (**19**) and interposed between said first window (**19**) and said first rim (**11**); said first window (**19**) being glued to said first rim (**11**) in the area of a first gluing bead (**70, 71**); and a second gasket (**65**) comolded on said second window (**26**) and glued to said first pillar (**14**) in the area of a second gluing bead (**75**), wherein said second gasket is interposed between said second window (**26**) and said first window (**19**), said first window (**19**) further comprising an upper edge (**61**) abutting against said first gasket (**60**) and glued to said first cross member (**12**) using said first gluing bead (**70**), a side edge (**62**) arranged so it abuts against said second gasket (**65**) and glued to said first pillar (**14**) using a third gluing bead (**71**), said second gasket being comolded on a side end edge (**66**) of said second window (**26**), said first gasket (**60**) and said second gasket (**65**) being arranged at corresponding adjacent sides of the corresponding first pillar (**14**).

2. The windshield according to claim 1, wherein said first rim (**10**) has a quadrangular shape and said second rim (**20**) has a triangular shape.

3. The windshield according to claim 1, wherein said third and fourth cross member (**22, 21**) of a said second and third rim (**20**) comprise respective first ends (**23**) connected to one another; each third cross member (**22**) comprising a respective second end opposite the respective first end (**23**) and connected to said first cross member (**12**); each fourth cross member (**21**) comprising a respective second end opposite the respective first end (**23**) and connected to said second cross member (**13**).

4. The windshield according to claim 1, wherein said first rim (**10**) is concave and/or said second and third rims (**20**) are concave.

5. The windshield according to claim 1, wherein said first cross member (12) has a concavity facing said second cross member (13) and said second cross member (13) has a concavity facing said first cross member (12).
6. The windshield according to claim 1, wherein at least one of said second and third window (26, 36) is made of tempered glass; and/or wherein said first window (19) is made of a pair of laminated glass plates, between which a film of plastic material is interposed.
7. A motor vehicle (1) comprising: a body (2) defining a passenger compartment (3); and a windshield (4) according to claim 1 and arranged at the front of said passenger compartment (3).
8. The motor vehicle according to claim 7, wherein the motor vehicle further comprises a plurality of accessories fixed to said first window (19) of said windshield (4).
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